



December 2025

(For MEC)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-



11. Which of the following best defines cloud computing?
 - A) Computing done entirely on local desktops
 - B) Delivery of computing services over the internet
 - C) Sharing files via email
 - D) Using only mobile devices for computation
12. Which of the following is *not* a benefit of cloud computing?
 - A) Scalability
 - B) On-demand resources
 - C) High upfront hardware cost
 - D) Flexibility
13. `cv2.approxPolyDP()` is used to:
 - A) Approximate a contour shape
 - B) Detect edges
 - C) Segment images
 - D) Detect blobs
14. Which method is used to detect car license plates in OpenCV?
 - A) Edge detection + Contour filtering
 - B) SIFT
 - C) ORB
 - D) FAST
15. Which tracker is suitable for object tracking?
 - A) KCF
 - B) Canny
 - C) SIFT
 - D) ORB
16. To capture video from a webcam, which function is used?
 - A) `cv2.VideoCapture`
 - B) `cv2.VideoWriter`
 - C) `cv2.imshow`
 - D) `cv2.Canny`
17. Which OpenCV class is used to detect QR codes?
 - A) QR Code Detector
 - B) Cascade Classifier
 - C) Tracker CSRT
 - D) Simple Blob Detector
18. What is the primary function of grippers in robotic systems?
 - A) Navigation
 - B) Object detection
 - C) Object manipulation
 - D) Data processing
19. Which sensor is commonly used for 3D mapping in drones?
 - A) Ultrasonic
 - B) Thermal
 - C) LIDAR
 - D) Infrared
20. Cobots are designed to work:
 - A) Independently
 - B) In hazardous zones
 - C) Alongside humans
 - D) Only in packaging

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the evolution from Industry 1.0 to Industry 5.0.	8	L*1	1	1
	b) A smart city authority reports that traffic congestion increased by 20% during peak hours compared to the previous year. The management wants an analytics report to plan future traffic signal optimization projects. Which method of analytics is better suited for this? What additional metrics could value-add this analytics?	8	L2	2	1
2.	a) Explain the temperature sensor and pressure sensor used in Industrial Processes.	8	L2	1	1
	b) A modern hospital wants to improve patient monitoring and safety using IoT. Currently the issues faced are - Patients with chronic heart conditions are not continuously monitored; Nurses				

manually track vitals, which delay critical response; Patients often face emergencies like sudden heart rate fluctuations. Management wants to propose an IoT-based healthcare system. For that, provide solutions for the following:

- i) Which types of IoT sensors would be deployed on patients?
- ii) What IoT architecture layers will process and use this data?

8 L2 2 1

- 3 a) Explain briefly about Intelligent Manufacturing.
- b) A manufacturing plant finds that machine downtime has increased by 18% in the last quarter, leading to reduced production efficiency. The operations team wants to know the reason behind this. What kind of analytics should be used for solving this? Suggest possible data sources for analysis.

8 L2 1 1

8 L2 2 1

Unit – II

- 4 a) A hospital is digitizing its operations and has the following requirements:

They want to store and manage patient medical records securely without investing in local data centers. Their IT team wants a platform to develop and deploy a custom patient management system. Doctors and staff need ready-to-use telemedicine, email, and document sharing tools for remote consultation.

- i) Which cloud service models (IaaS, PaaS, SaaS) are suitable for each of the above requirements?

- ii) Give one example provider/tool for each case.

10 L3 3 1

- b) Explain the procedure for finding Waldo in an image using template matching or feature detection.

3 L1 4 1

- c) Describe contour detection and how to identify contours by shape in OpenCV.

3 L2 4 1

- 5 a) A hospital wants to analyze patient admissions and discharge trends using Power BI. Data comes from multiple sources: SQL database (Patient_Records), CSV files (Lab_Results.csv), and Excel files (Staff_Assignments.xlsx).

- i) How would you import data from these different sources into Power BI?

- ii) What preparation steps are needed before building visualizations?

10 L3 3 1

- b) Explain how to detect a car license plate in an image using OpenCV.

3 L2 4 1

- c) Describe the steps involved in face detection in an image using OpenCV.

3 L2 4 1

- 6 a) Explain object tracking techniques in OpenCV with an example scenario.

4 L2 4 1

- b) Describe how to connect to a USB or laptop camera and record video using OpenCV.

4 L2 4 1

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|--|---|----|---|---|
| c) Explain QR-Code detection in an image or video using OpenCV. | 4 | L2 | 4 | 1 |
| d) Discuss feature detection algorithms (SIFT, SURF, FAST, BRIEF, ORB) and their differences in performance. | 4 | L1 | 4 | 1 |

Unit – III

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|---|---|----|---|---|
| 7. a) Explain the role of Robots and Drones in Advanced Manufacturing. | 8 | L2 | 5 | 1 |
| b) Describe the components and benefits of vision-assisted robots in packaging. | 8 | L1 | 5 | 1 |
| 8 a) Discuss the role of drones in defect detection. | 8 | L1 | 5 | 1 |
| b) Describe the working of a Vision Guided Robotic Assembly Line. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
